

TEST CASE ID	SECTION	TEST DESCRIPTION	PRECONDITIONS	Test Data	TEST STEPS	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC001	Username Field	Check if username field is visible	-	-	Load page and verify presence of username field	Username field should be visible		
TC002	Username Field	Check if username field is editable	-	testuser	Type in the username field	Field should accept input		
TC003	Username Field	Check placeholder text	-	-	Observe placeholder text in username field	Placeholder should be "Username"		
TC004	Username Field	Input special characters in username	-	@user123!	Enter special characters in username field	Field should accept input		
TC005	Username Field	Input max length username	-	50+ chars	Enter long username string	Should accept or trim input		
TC006	Username Field	Leave username empty	-	-	Leave username blank and try to sign in	Show error for empty username		
TC007	Username Field	Input username with leading/trailing spaces	-	" user1 "	Enter username with spaces	Should trim spaces		
TC008	Username Field	Input with Unicode characters	-	用户名	Enter Unicode string in username	Should accept or validate		
TC009	Username Field	Autofill check	Browser autofill	Saved credentials	Click into username field	Autofill should appear		
TC010	Username Field	Case sensitivity test	Case-sensitive system	User1 / user1	Try login with case differences	Should treat differently		
TC011	Username Field	Copy-paste into username	-	Copied from notepad	Paste username into field	Should accept paste		
TC012	Username Field	Backspace works in username	-	testuser	Press backspace key in username field	Characters should be deleted		
TC013	Username Field	Login with only username	-	testuser / -	Enter username only, click login	Show password required error		
TC014	Password Field	Check if password field is visible	-	-	Load page and verify password field	Password field should be visible		
TC015	Password Field	Check if password is masked	-	pass123	Enter password	Characters should be masked		
TC016	Password Field	Check eye icon toggle	-	pass123	Click eye icon to reveal/hide	Password visibility should toggle		
TC017	Password Field	Password field editable	-	pass123	Type into password field	Field should accept input		
TC018	Password Field	Placeholder text	-	-	Check placeholder in password field	Should be "Password"		
TC019	Password Field	Input special characters in password	-	P@ssw0rd!	Enter password with symbols	Should accept input		
TC020	Password Field	Input long password	-	100+ chars	Enter very long password	Should be handled properly		
TC021	Password Field	Backspace works in password	-	pass123	Press backspace in password field	Should delete characters		
TC022	Password Field	Copy-paste into password	-	Copied text	Paste into password field	Paste should work		
TC023	Password Field	Unicode characters in password	-	你好123	Enter Unicode password	Should accept or validate		
TC024	Password Field	Login with only password	-	- / pass123	Leave username empty and click login	Show username required error		
TC025	Password Field	Password too short	Enforced min length	1 char	Enter short password and click login	Error for password too short		
TC026	Password Field	Toggle eye icon twice	-	pass123	Toggle twice	Should return to masked state		
TC027	Sign In Button	Sign In button is visible	-	-	Check visibility of button	Should be visible		
TC028	Sign In Button	Sign In button is clickable	-	-	Try clicking the button	Should respond on click		
TC029	Sign In Button	Valid login	Valid credentials	user1 / pass123	Fill and click Sign In	Should login successfully		
TC030	Sign In Button	Both fields empty	-	-	Click Sign In with empty fields	Show error message		
TC031	Sign In Button	Invalid credentials	-	wrong / wrongpass	Try login with wrong credentials	Show invalid login error		
TC032	Sign In Button	Spaces in both fields	-	" " / " "	Try with only spaces	Show input validation error		
TC033	Sign In Button	Double-click sign-in	-	valid creds	Double-click button	Should not double-submit		
TC034	Sign In Button	SQL Injection attempt	-	OR 1=1 -- / test	Try SQL injection payload	Should not bypass login		
TC035	Sign In Button	Enter key triggers login	Fields filled	valid creds	Press Enter instead of button	Should trigger login		
TC036	Sign In Button	Long username + password	-	100+ char creds	Try very long input	Should not break UI		
TC037	Sign In Button	Sign In button default enabled	-	-	Load page and observe	Button should be clickable		
TC038	Sign In Button	Button color consistency	-	-	Observe button color	Should match theme		
TC039	Sign In Button	Show loader on login	Optional	valid creds	Click login and watch for spinner	Spinner or feedback should show		
TC040	Sign In Button	Disabled on empty input	Optional	-	Leave fields empty	Button may be disabled		
TC041	Sign Up Link	Check visibility of sign-up link	-	-	Look for "Sign Up"	Link should be visible		
TC042	Sign Up Link	Redirect to sign-up page	-	-	Click on "Sign Up" link	Should go to sign-up page		
TC043	UI	Welcome message is visible	-	-	Look for welcome message	Should say "Welcome to Optimized RAG"		
TC044	UI	Font consistency check	-	-	Observe font types	Fonts should match across inputs		
TC045	UI	Input field alignment	-	-	Check if fields align properly	Should be aligned		
TC046	UI	Button alignment	-	-	Check if button aligns properly	Should be aligned		
TC047	UI	Mobile responsiveness	-	-	View in mobile resolution	Should resize properly		
TC048	UI	Tablet responsiveness	-	-	View in tablet resolution	Should resize properly		
TC049	UI	Desktop responsiveness	-	-	View in desktop resolution	Should scale properly		
TC050	UI	Field spacing	-	-	Check vertical spacing	Should have consistent spacing		
TC051	UI	Placeholder styling	-	-	Observe placeholder color and font	Should match theme		
TC052	UI	Border color on focus	-	-	Click input field	Should change border color		
TC053	UI	Error message visibility	-	-	Submit with invalid input	Error should be shown		

TC054	UI	Error message styling	-	-	Trigger error	Message should be styled		
TC055	UI	Page loads within 2 seconds	Good internet	-	Load page	Should load quickly		
TC056	UI	Contrast ratio compliance	-	-	Check contrast for text	Should pass accessibility		
TC057	UI	Focus indicator visible	-	-	Tab through inputs	Should show outline		
TC058	UI	Keyboard navigation	-	-	Use Tab/Shift+Tab and Enter	Should fully navigate		
TC059	UI	Tab order is logical	-	-	Use Tab key through inputs	Should follow input order		
TC060	UI	No horizontal scroll	-	-	Check for horizontal scroll	Should not exist		
TC061	UI	Page title in browser tab	-	-	Check browser tab title	Should be meaningful		
TC062	UI	Page has favicon	-	-	Check if favicon appears	Favicon should be present		
TC063	Validation	Show error on empty username	-	-	Leave username blank, click login	Should show error message		
TC064	Validation	Show error on empty password	-	user1 / -	Leave password blank, click login	Should show error message		
TC065	Validation	Error disappears after correction	Trigger error first	valid creds	Enter invalid data → correct it → click login	Error should disappear		
TC066	Validation	Error message text accuracy	-	-	Trigger error intentionally	Message should match expected format		
TC067	Validation	Show error on invalid format username	Format validation enabled	bad@format	Enter bad format username, click login	Should show error		
TC068	Validation	Error message styling (color, font)	Error appears	-	Trigger any error	Error message should match design		
TC069	Validation	Server-side error handling	Backend returns 500	valid creds	Simulate server failure	Show appropriate message		
TC070	Validation	Rate limit or brute force block	Multiple invalid attempts	-	Fail login 5+ times	Should block or alert		
TC071	Validation	Error persists after page refresh	With sessionStorage	-	Trigger error → refresh page	Error should or should not persist		
TC072	Validation	Form auto-submit prevention	Form with JS logic	-	Trigger auto-submit	Should block/handle gracefully		
TC073	Validation	Non-field error rendering	Backend sends general msg	-	Submit with backend error	Message should display on top		
TC074	Validation	Password error shown only if password wrong	Wrong password only	user / wrongpass	Enter valid user, wrong pass	Should specify password error		
TC075	Validation	Invalid inputs don't clear on refresh	Manual refresh	bad creds	Enter and refresh page	Fields should reset		
TC076	Browser Support	Test on Chrome (latest)	-	valid creds	Load and test login	Should work fine		
TC077	Browser Support	Test on Firefox	-	valid creds	Load and test login	Should work fine		
TC078	Browser Support	Test on Safari	Mac or iOS	valid creds	Load and test login	Should work fine		
TC079	Browser Support	Test on Microsoft Edge	-	valid creds	Load and test login	Should work fine		
TC080	Browser Support	Test on Brave	-	valid creds	Load and test login	Should work fine		
TC081	Browser Support	Zoom in/out (browser zoom)	-	-	Zoom to 50%, 150%	Page should remain usable		
TC082	Browser Support	Incognito mode test	-	valid creds	Open incognito and login	Should behave same		
TC083	Browser Support	Disable JS and load	JS disabled	-	Load page	Should show fallback or warning		
TC084	Browser Support	Cookie disabled	Cookies disabled	-	Load and attempt login	Should warn or fail safely		
TC085	Browser Support	Test on mobile Chrome	Android	valid creds	Load on mobile browser	Should be responsive		
TC086	MSAL Login	Valid login using popup	MSAL instance initialized	Valid Microsoft account	1. Click Microsoft login button 2. Popup appears 3. Enter valid credentials 4. Submit	User is logged in and token is acquired	TRUE	
TC087	MSAL Login	Popup login fails, fallback to redirect	MSAL popup blocked or error thrown	Valid Microsoft account	1. Simulate popup block/error 2. Fallback to redirect	User is redirected to Microsoft login and logged in		
TC088	MSAL Login	Silent token acquisition succeeds	User logged in	Valid session token	1. Perform login 2. Call acquireTokenSilent	Access token is acquired silently		
TC089	MSAL Login	Silent token acquisition fails, fallback to popup	Session expired	Valid credentials	1. Expire session 2. Try silent token 3. Fallback to popup	Token acquired via popup		
TC090	MSAL Login	Access Graph API with token	Valid token acquired	GraphMe endpoint	1. Fetch user profile with access token	User details received		
TC091	MSAL Login	Invalid credentials popup login	None	Invalid credentials	1. Enter wrong credentials in popup 2. Redirect to Microsoft login	Error shown, login failed		
TC092	MSAL Login	Invalid credentials redirect login	None	Invalid credentials	1. Simulate popup block 2. Redirect to Microsoft login 3. Enter wrong creds	Error shown, login failed		
TC093	MSAL Login	Cancel login popup	None	Cancel login window	1. Open popup 2. Close window without submitting	Login aborted, error shown		
TC094	MSAL Login	Login popup closed immediately	None	Close popup	1. Immediately close popup on opening	Login fails, appropriate error handled		
TC095	MSAL Token	Expired access token re-acquisition	Token expired	Valid session	1. Expire token 2. Call acquireTokenSilent	Token reacquired via popup/silent		

TC096	MSAL Token	Graph API returns error	Invalid token	Invalid token	1. Use malformed/expired token 2. Call Graph API	Graph API returns 401, error handled
TC097	MSAL Login	Network failure during popup login	None	Simulate network loss	1. Disconnect network 2. Try login	Network error handled gracefully
TC098	MSAL Login	Corrupt redirect URI	Wrong redirect URI in Azure config	Mismatched URI	1. Click login 2. Azure can't redirect	Login fails, hash_empty_error shown
TC099	MSAL Login	Redirect URI not in Azure portal	Missing redirect URI	Unregistered URI	1. Use invalid redirect URI 1. Trigger login	Azure login fails with configuration error
TC100	MSAL Login	Multiple accounts signed in	Two Microsoft accounts in browser	Account selection prompt	2. Select different account 1. Complete login	Prompt shown for account selection
TC101	MSAL Login	Access token not returned	Azure misconfigured	Incorrect API permissions	2. Access token not received	Error shown for token acquisition
TC102	MSAL Flow	No account returned after login	Empty login result	Mock login response	1. Simulate empty account in login response 1. Complete login	Login fails, handled gracefully
TC103	MSAL Flow	handleSuccessfulLogin gets correct data	Valid login	MS Graph returns data	2. Call handleSuccessfulLogin	Receives username, token, welcome message
TC104	MSAL Error Handling	Catch all MSAL errors	Simulate unknown error	Throw unknown error	1. Simulate throw in loginPopup	Catch block executed, notification shown
TC105	MSAL UI Feedback	Loading spinner active during login	Trigger login	Valid credentials	1. Click login button 1. Simulate error	Spinner appears until login completes
TC106	MSAL UI Feedback	Error toast shown on login failure	Login fails	Any MSAL error	2. Catch and show toast	Toast with "Login Failed" appears
TC107	MSAL Login	Token sent in Authorization header	Valid access token	Call to Graph endpoint	1. Inspect request headers	Authorization: Bearer <token> is set
TC108					1. Login once 2. Refresh page 3. Login again	Token is reused silently
TC109	MSAL Session	Token reused on second login attempt	Token stored in MSAL	Repeat login		
TC110	MSAL Config	loginRequest passed to loginPopup	loginRequest defined	Check parameters	1. Inspect loginRequest in popup call	Scope and redirect URI matches config
TC111	MSAL Login	loginRedirect throws error	Invalid config or no redirect handler	loginRedirect called	1. Simulate error in loginRedirect 1. Open same app in two tabs	Error shown and caught
TC112	MSAL Behavior	Multiple simultaneous login attempts	Two tabs	Click login in both tabs	2. Click login simultaneously	One succeeds, other shows error
TC113	MSAL Redirect URI	Redirect URI must match exactly	Slight mismatch	Typo in URI	1. Configure wrong URI (e.g., missing 1. Copy token	Login fails with URI mismatch error
TC114	MSAL Security	Token cannot be used on other client	Acquire token in one app	Use in another	2. Try in another MSAL client 1. Perform login	Token rejected
TC115	MSAL Storage	MSAL stores session in localStorage/sessionStorage	MSAL config set	Inspect browser storage	2. Inspect local/sessionStorage	MSAL session keys found
TC116	MSAL Multi Tenancy	Login works with multiple tenant IDs	Multi-tenant app	Tenant A and B credentials	1. Login with different tenant users	Both logins succeed if app supports tenants